

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

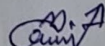
Annexure A

Analytical Problem Solving

Code of Conduct:

I Bhavyanka . G / 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Level - 4


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1. For a Tic-Tac-Toe playing agent, define its PEAS description and characterize the environment (is it fully observable? deterministic? episodic? single or multi-agent?). Explain why this environment is classified as adversarial.
2. A warehouse company wants to deploy robots to pick and transport goods to packing stations. Identify which agent architecture (simple reflex, model-based, goal-based, utility-based, or learning agent) is most appropriate, and justify your choice with reasons specific to the warehouse environment.
3. Consider the map of Romania (Arad to Bucharest) as a graph of cities connected by roads with distances. Trace the order in which cities are expanded using Breadth-First Search and then Depth-First Search, starting from Arad. Compare the two traces in terms of completeness and optimality.

DATE: 25/07/26

CSA1714 ARTIFICIAL INTELLIGENCE

NAME : PRIYANKA . G

CO 1 ASSESSMENT TOOL 1

REG. NO: 192524124

ANALYTICAL PROBLEM
SOLVING

1) TIC-TAC-TOE PLAYING AGENT

(a) PEAS DESCRIPTION

Component	Description
Performance Measure	Win the game, avoid losing, maximize number of wins, minimize draws/losses.
Environment	3x3 Tic-Tac-Toe board, opponent player, game rules.
Actuators	Place x or O in an empty cell on board.
Sensors	Observe the current board state and opponent's move.

(b) Environment Characters

Property	Classification	Reason
Fully observable	Yes	The entire board is visible to both players at all times.
Deterministic	Yes	Every move leads to a predictable board state.
Episodic or sequential	Sequential	Each move affects all future moves.
Static or Dynamic	Static	The board changes only when a player makes a move.
Discrete or Continuous	Discrete	Limited number of board positions and legal actions.
Single or Multi Agent	Multi-Agent	Two players compete against each other.

(c) Environment Adversarial. (multi-agent)

The Tic-Tac-Toe environment is adversarial because two intelligent agents have conflicting objectives.

- One player's gain is the other player's loss.
 - Each player attempts to block the opponent's winning strategy and the agent must predict the opponent's possible moves before selecting their own move.
 - Decision making depends not only on the current board but also on anticipating future opponent actions.
- ⇒ Hence, Tic-Tac-Toe is a competitive multi-agent environment.

Therefore, Tic-Tac-Toe is a fully observable, deterministic, sequential environment based on PEAS framework.

2) WAREHOUSE ROBOT AGENT.

→ The most appropriate agent architecture : Learning Agent.

→ Justification :

A warehouse robot operates in a changing and dynamic environment where it must efficiently transport goods while avoiding obstacles and optimizing travel paths.

→ Reasons :

(i) Adapts to dynamic environment :

Warehouse layout, worker movements and obstacles change frequently. A 'Learning Agent' improves its performance based on previous experiences.

(ii) Learns efficient routes :

The robot can learn shortest paths, traffic patterns and frequently used shelves which reduce travel time.

(iii) Improves picking accuracy:

The robot learns object locations, grasping techniques, handling fragile items leading to fewer errors.

(iv) Handles Unexpected Situations:

Learning agents adapts when shelves are moved, paths are blocked, inventory changes without requiring manual reprogramming.

(v) Optimizes Overall Performance:

The robot continuously improves delivered speed, battery usage, task scheduling, collision avoidance thereby increasing warehouse productivity.

→ Conclusion:

A Learning agent is the most suitable architecture because it continuously learns from experience, adapts to changing warehouse conditions and optimizes efficiency over time.

3) ROMANIA MAP SEARCH (ARAD → BUCHAREST)

Assume the standard Romania map used in Artificial Intelligence.

(a) Breadth-First search (BFS)

BFS explores cities level by level.

> Expansion Order:

Arad → Zerind → Sibiu → Timisoara → Oradea →
Fagaras → Rimnicu Vilcea → Lugoj → Bucharest (Goal Found)

> Path Found:

Arad → Sibiu → Fagaras → Bucharest.

(b) Depth-First Search (DFS)

DFS explores one branch completely before backtracking.

> Expansion Order:

1. Arad
2. Zerind
3. Oradea
4. Sibiu
5. Fagaras
6. Bucharest (Goal Found).

> Path Found:

Arad → Zerind → Oradea → Sibiu → Fagaras → Bucharest.

~ DFS order depends on the order in which neighboring cities are visited.

(c) Comparison of BFS and DFS

Feature	BFS	DFS
Search Strategy	Explores level by level	Explores one branch deeply first.
Data Structure	Queue (FIFO)	Stack (LIFO)
Completeness	Complete	Not always complete (in infinite spaces)
Optimality	Optimal when all steps costs are equal	Not optimal.
Memory	High	Low.
Time	Higher	Explores long unnecessary paths.

→ Conclusion

» BFS: • Explores all neighboring cities first
• Guarantees finding the shortest path when all edges have equal cost and uses more memory.

» DFS: • Explores one path as deeply as possible before backtracking.
• Requires less memory.
• Doesn't guarantee the shortest path & may explore unnecessary branches first.